EXP-1 IMAGE CAPTIONING USING CNN-RNN ARCHITECTURE

AIM:

To generate ~~meanig~~ meaningful captions for images using a pretrained image encoder and text-generation decoder

SOFTWARE REQUIREMENTS

* Python 3.x
* Pytorch
* Hugging face Transformers
* PIL
* Pretrained ViT-GPT-2 model

ALGORITHM:

* Import the required libraries
* Load a pretrained image-captioning model
* Load the image processor and tokenizer
* Read the input image
* Convert the image into the required tensor format
* Generate a caption using beam search
* Decode and display the generated caption

PROGRAM:

```
import torch
from PIL import Image
from transformers import (
    VisionEncoderDecoderModel,
    ViTImageProcessor,
    AutoTokenizer,
)
model_name = "nlpconnect/vit-gpt2-image-captioning"
```

EXPECTED OUTPUT:

Generated Caption: A person standing near a ....

```
device = torch.device("cuda if torch.cuda.is_available() else "cpu"
    )
model = VisionEncoderDecoderModel.from_pretrained(model name)
        .to(device)
processor = ViTImageProcessor.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)
model.eval()

def generate_caption(image_path):~~convert("RGB")~~
    image = Image.open(image_path).convert("RGB")
    pixels = processor(
        images = image,
        return_tensor="pt"
    ).pixel_values.to(device)

    ~~with~~
    return tokenizer.decode(
        output[0],
        skip_special_tokens=True
    )
image_path = "download.jpeg"
print("Generated caption:", generate_caption(image_path))
```

RESULT:

The pretrained image-captioning model was implemented to generate textual description of images using visual feature generation and language generation.